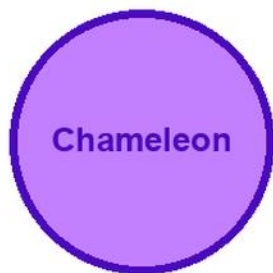


KITABU QUEST

Standard V

SAYANSI



Cover scene

learners observing plants and simple experiments with a chameleon mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

V

Title Page

KITABU QUEST Standard V Sayansi

Tanzania TIE-BASIC learner book

Mascot: chameleon

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Tanzania learners.

How To Use This Book

This book helps Standard V learners study Sayansi one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Chunguza, uliza maswali, jaribu kwa usalama, andika ushahidi, na eleza matokeo.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Human Body Systems And Health
2. Plants Growth And Reproduction
3. Soil Water And Agriculture
4. Energy Electricity And Magnetism
5. Matter Mixtures And Changes
6. Weather Climate And Environment
7. Technology Design And Simple Machines
8. Science Investigation Review

As you finish each topic, return to the success criteria and correct one answer before moving on.

Human Body Systems And Health: Lesson Opener

Learning focus: Human Body Systems And Health

Country link: a safe Tanzania observation, model, data table, or investigation for Human Body Systems And Health.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain human body systems and health using evidence

Inquiry questions:

- How can human body systems and health help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human Body Systems And Health: Learn

Human Body Systems And Health is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Human Body Systems And Health
- Human
- Body
- Systems
- Health
- concept
- observation
- evidence

Human Body Systems And Health: Worked Example

Investigation model for Human Body Systems And Health: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Human Body Systems And Health: Vocabulary And Study Skill

Use these words and study moves before you practise Human Body Systems And Health. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Human Body Systems And Health
- Human
- Body
- Systems
- Health
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Human Body Systems And Health without a clear example, method, or evidence.

Human Body Systems And Health: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Human Body Systems And Health.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Human Body Systems And Health: Practice And Correction

Practice for Human Body Systems And Health:

Main outcome: investigate and explain human body systems and health using evidence.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Human Body Systems And Health: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Human Body Systems And Health.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human Body Systems And Health: Outcome Check

Outcome check for Human Body Systems And Health: Investigate and explain human body systems and health using evidence.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants Growth And Reproduction: Lesson Opener

Learning focus: Plants Growth And Reproduction

Country link: a safe Tanzania observation, model, data table, or investigation for Plants Growth And Reproduction.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain plants growth and reproduction using evidence

Inquiry questions:

- How can plants growth and reproduction help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants Growth And Reproduction: Learn

Plants Growth And Reproduction teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Plants Growth And Reproduction
- Plants
- Growth
- Reproduction
- concept
- observation
- evidence
- safety

Plants Growth And Reproduction: Worked Example

Worked example for Plants Growth And Reproduction: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Plants Growth And Reproduction: Vocabulary And Study Skill

Use these words and study moves before you practise Plants Growth And Reproduction. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Plants Growth And Reproduction
- Plants
- Growth
- Reproduction
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Plants Growth And Reproduction without a clear example, method, or evidence.

Plants Growth And Reproduction: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Plants Growth And Reproduction.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Plants Growth And Reproduction: Practice And Correction

Practice for Plants Growth And Reproduction:

Main outcome: investigate and explain plants growth and reproduction using evidence.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Plants Growth And Reproduction: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Plants Growth And Reproduction.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants Growth And Reproduction: Outcome Check

Outcome check for Plants Growth And Reproduction: Investigate and explain plants growth and reproduction using evidence.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Soil Water And Agriculture: Lesson Opener

Learning focus: Soil Water And Agriculture

Country link: a safe Tanzania observation, model, data table, or investigation for Soil Water And Agriculture.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain soil water and agriculture using evidence

Inquiry questions:

- How can soil water and agriculture help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Soil Water And Agriculture: Learn

Soil Water And Agriculture explains soil as a living natural resource that supports plants, animals, and people.

Core idea:

- Soil is made from weathered rock, humus, air, water, and living organisms.
- Sandy, clay, and loam soils feel different, hold water differently, and suit different uses.
- Healthy soil supports plant roots, stores water, and provides nutrients.
- Soil can be damaged by erosion, burning, pollution, and careless farming.
- Learners can compare soil samples safely by colour, texture, water retention, and visible organic matter.

Key words:

- Soil Water And Agriculture
- Soil
- Water
- Agriculture
- concept
- observation
- evidence
- safety

Soil Water And Agriculture: Worked Example

Investigation model for Soil Water And Agriculture: Compare two soil samples safely.

1. Collect two small teacher-approved soil samples, such as garden soil and sandy soil.
2. Observe colour and texture without putting soil near the mouth or eyes.
3. Add a small amount of water and compare how each sample holds water.
4. Record results in a table: sample, colour, texture, water retention, visible organic matter.
5. Explain which sample may support plant growth better and why.

Conclusion: soil properties affect plant growth and land use.

Soil Water And Agriculture: Vocabulary And Study Skill

Use these words and study moves before you practise Soil Water And Agriculture. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Soil Water And Agriculture
- Soil
- Water
- Agriculture
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Soil Water And Agriculture without a clear example, method, or evidence.

Soil Water And Agriculture: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Soil Water And Agriculture.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Soil Water And Agriculture: Practice And Correction

Practice for Soil Water And Agriculture:

Main outcome: investigate and explain soil water and agriculture using evidence.

1. Compare two safe soil samples by colour, texture, and water retention.
2. Record observations in a table.
3. Explain which soil may support plant growth better and why.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Soil Water And Agriculture: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Soil Water And Agriculture.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Soil Water And Agriculture: Outcome Check

Outcome check for Soil Water And Agriculture: Investigate and explain soil water and agriculture using evidence.

1. Name two soil components.
2. Compare two soil properties.
3. Explain one way to protect soil from damage.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Energy Electricity And Magnetism: Lesson Opener

Learning focus: Energy Electricity And Magnetism

Country link: a safe Tanzania observation, model, data table, or investigation for Energy Electricity And Magnetism.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain energy electricity and magnetism using evidence

Inquiry questions:

- How can energy electricity and magnetism help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Energy Electricity And Magnetism: Learn

Energy Electricity And Magnetism connects work, power, and energy changes.

Core idea:

- Work is done when a force moves an object through a distance.
- Energy is the ability to do work.
- Power describes how quickly work is done or energy is transferred.
- Energy can change form, such as chemical to movement, electrical to light, or potential to kinetic.

Key words:

- Energy Electricity And Magnetism
- Energy
- Electricity
- Magnetism
- concept
- observation
- evidence
- safety

Energy Electricity And Magnetism: Worked Example

Worked example for Energy Electricity And Magnetism: A learner lifts a book from the floor to a desk.

1. The learner applies an upward force.
2. The book moves upward through a distance.
3. Work is done on the book.
4. The book gains gravitational potential energy.

Conclusion: work can transfer energy to an object.

Energy Electricity And Magnetism: Vocabulary And Study Skill

Use these words and study moves before you practise Energy Electricity And Magnetism. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Energy Electricity And Magnetism
- Energy
- Electricity
- Magnetism
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Energy Electricity And Magnetism without a clear example, method, or evidence.

Energy Electricity And Magnetism: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Energy Electricity And Magnetism.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Energy Electricity And Magnetism: Practice And Correction

Practice for Energy Electricity And Magnetism:

Main outcome: investigate and explain energy electricity and magnetism using evidence.

1. Give two examples of energy transformation.
2. Explain when work is done on an object.
3. Compare work and power in one sentence.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Energy Electricity And Magnetism: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Energy Electricity And Magnetism.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Energy Electricity And Magnetism: Outcome Check

Outcome check for Energy Electricity And Magnetism: Investigate and explain energy electricity and magnetism using evidence.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Matter Mixtures And Changes: Lesson Opener

Learning focus: Matter Mixtures And Changes

Country link: a safe Tanzania observation, model, data table, or investigation for Matter Mixtures And Changes.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain matter mixtures and changes using evidence

Inquiry questions:

- How can matter mixtures and changes help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Matter Mixtures And Changes: Learn

Matter Mixtures And Changes is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Matter Mixtures And Changes
- Matter
- Mixtures
- Changes
- concept
- observation
- evidence
- safety

Matter Mixtures And Changes: Worked Example

Investigation model for Matter Mixtures And Changes: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Matter Mixtures And Changes: Vocabulary And Study Skill

Use these words and study moves before you practise Matter Mixtures And Changes. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Matter Mixtures And Changes
- Matter
- Mixtures
- Changes
- concept

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Matter Mixtures And Changes without a clear example, method, or evidence.

Matter Mixtures And Changes: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Matter Mixtures And Changes.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Matter Mixtures And Changes: Practice And Correction

Practice for Matter Mixtures And Changes:

Main outcome: investigate and explain matter mixtures and changes using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Matter Mixtures And Changes: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Matter Mixtures And Changes.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Matter Mixtures And Changes: Outcome Check

Outcome check for Matter Mixtures And Changes: Investigate and explain matter mixtures and changes using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Weather Climate And Environment: Lesson Opener

Learning focus: Weather Climate And Environment

Country link: a safe Tanzania observation, model, data table, or investigation for Weather Climate And Environment.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain weather climate and environment using evidence

Inquiry questions:

- How can weather climate and environment help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Weather Climate And Environment: Learn

Weather Climate And Environment is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Weather Climate And Environment
- Weather
- Climate
- Environment
- concept
- observation
- evidence
- safety

Weather Climate And Environment: Worked Example

Investigation model for Weather Climate And Environment: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Weather Climate And Environment: Vocabulary And Study Skill

Use these words and study moves before you practise Weather Climate And Environment. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Weather Climate And Environment
- Weather
- Climate
- Environment
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Weather Climate And Environment without a clear example, method, or evidence.

Weather Climate And Environment: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Weather Climate And Environment.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Weather Climate And Environment: Practice And Correction

Practice for Weather Climate And Environment:

Main outcome: investigate and explain weather climate and environment using evidence.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Weather Climate And Environment: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Weather Climate And Environment.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Weather Climate And Environment: Outcome Check

Outcome check for Weather Climate And Environment: Investigate and explain weather climate and environment using evidence.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Technology Design And Simple Machines: Lesson Opener

Learning focus: Technology Design And Simple Machines

Country link: a safe Tanzania observation, model, data table, or investigation for Technology Design And Simple Machines.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain technology design and simple machines using evidence

Inquiry questions:

- How can technology design and simple machines help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Technology Design And Simple Machines: Learn

Technology Design And Simple Machines shows how tools make work easier.

Core idea:

- A simple machine changes the size or direction of a force.
- Levers, pulleys, wheels and axles, inclined planes, wedges, and screws are simple machines.
- A machine does not remove work; it helps you use force more conveniently.
- Compare effort force, load, and distance moved.

Key words:

- Technology Design And Simple Machines
- Technology
- Design
- Simple
- Machines
- concept
- observation
- evidence

Technology Design And Simple Machines: Worked Example

Investigation model for Technology Design And Simple Machines: Use a ruler as a lever.

1. Place a small object as the load.
2. Put a pencil under the ruler as the pivot.
3. Press the other end gently.
4. Move the pivot and compare the effort needed.

Conclusion: the position of the pivot changes how a lever helps lift a load.

Technology Design And Simple Machines: Vocabulary And Study Skill

Use these words and study moves before you practise Technology Design And Simple Machines. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Technology Design And Simple Machines
- Technology
- Design
- Simple
- Machines
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Technology Design And Simple Machines without a clear example, method, or evidence.

Technology Design And Simple Machines: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Technology Design And Simple Machines.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Technology Design And Simple Machines: Practice And Correction

Practice for Technology Design And Simple Machines:

Main outcome: investigate and explain technology design and simple machines using evidence.

1. Name three simple machines.
2. Identify the load, effort, and pivot in a lever.
3. Explain how changing the pivot changes the effort.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Technology Design And Simple Machines: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Technology Design And Simple Machines.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Technology Design And Simple Machines: Outcome Check

Outcome check for Technology Design And Simple Machines: Investigate and explain technology design and simple machines using evidence.

1. Name one simple machine.
2. Identify load, effort, and pivot or contact point.
3. Explain how it makes work easier.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Science Investigation Review: Lesson Opener

Learning focus: Science Investigation Review

Country link: a safe Tanzania observation, model, data table, or investigation for Science Investigation Review.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain science investigation review using evidence

Inquiry questions:

- How can science investigation review help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Science Investigation Review: Learn

Science Investigation Review is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Science Investigation Review
- Science
- Investigation
- Review
- concept
- observation
- evidence
- safety

Science Investigation Review: Worked Example

Investigation model for Science Investigation Review: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Science Investigation Review: Vocabulary And Study Skill

Use these words and study moves before you practise Science Investigation Review. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Science Investigation Review
- Science
- Investigation
- Review
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Science Investigation Review without a clear example, method, or evidence.

Science Investigation Review: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Science Investigation Review.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Science Investigation Review: Practice And Correction

Practice for Science Investigation Review:

Main outcome: investigate and explain science investigation review using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Science Investigation Review: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Science Investigation Review.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Science Investigation Review: Outcome Check

Outcome check for Science Investigation Review: Investigate and explain science investigation review using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human Body Systems And Health: Review Clinic 1

Use this review clinic to strengthen Sayansi without copying earlier answers.

1. Choose one idea from Human Body Systems And Health that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Plants Growth And Reproduction: Review Clinic 2

Use this review clinic to strengthen Sayansi without copying earlier answers.

1. Choose one idea from Plants Growth And Reproduction that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Soil Water And Agriculture: Review Clinic 3

Use this review clinic to strengthen Sayansi without copying earlier answers.

1. Choose one idea from Soil Water And Agriculture that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Energy Electricity And Magnetism: Review Clinic 4

Use this review clinic to strengthen Sayansi without copying earlier answers.

1. Choose one idea from Energy Electricity And Magnetism that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Human Body Systems And Health: Application Workshop 1

Application workshop 1 for Human Body Systems And Health.

Grade move: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Plants Growth And Reproduction: Application Workshop 2

Application workshop 2 for Plants Growth And Reproduction.

Grade move: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Soil Water And Agriculture: Application Workshop 3

Application workshop 3 for Soil Water And Agriculture.

Grade move: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Energy Electricity And Magnetism: Application Workshop 4

Application workshop 4 for Energy Electricity And Magnetism.

Grade move: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Matter Mixtures And Changes: Application Workshop 5

Application workshop 5 for Matter Mixtures And Changes.

Grade move: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Weather Climate And Environment: Application Workshop 6

Application workshop 6 for Weather Climate And Environment.

Grade move: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Technology Design And Simple Machines: Application Workshop 7

Application workshop 7 for Technology Design And Simple Machines.

Grade move: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Science Investigation Review: Application Workshop 8

Application workshop 8 for Science Investigation Review.

Grade move: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Human Body Systems And Health: Application Workshop 9

Application workshop 9 for Human Body Systems And Health.

Grade move: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Plants Growth And Reproduction: Application Workshop 10

Application workshop 10 for Plants Growth And Reproduction.

Grade move: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Glossary

Use this glossary to revise important words in Sayansi.

- Human Body Systems And Health: Explain this term using an example from the book, your school, or your community.
- Plants Growth And Reproduction: Explain this term using an example from the book, your school, or your community.
- Soil Water And Agriculture: Explain this term using an example from the book, your school, or your community.
- Energy Electricity And Magnetism: Explain this term using an example from the book, your school, or your community.
- Matter Mixtures And Changes: Explain this term using an example from the book, your school, or your community.
- Weather Climate And Environment: Explain this term using an example from the book, your school, or your community.
- Technology Design And Simple Machines: Explain this term using an example from the book, your school, or your community.
- Science Investigation Review: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Sayansi answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Sayansi that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.